TEXORA · SESSIONAL NOTES · STUDY SHEET

Practical Study on Industrial Sewing Machine

Identifying every working part of the single-needle lockstitch and overlock machines, tracing the stitch-forming cycle, and verifying machine performance with measured speed, stitch density and thread-consumption calculations.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 1 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to identify the main parts of the industrial single-needle lockstitch (SNLS) machine and the overlock machine, to explain the function of each part in the stitch-forming cycle, and to verify the machine's working performance by measuring speed, stitch density and thread consumption on a sewn specimen.

This study is the foundation of the sessional: every later study — garment production, line balancing, the case study — assumes the student can look at a machine, name its parts, and say what each one does. The threading and adjustment detail built on this study is practised in TE-3108 and is used here without repetition.

2. Materials, machines and specimens

Machines: one SNLS lockstitch machine (stitch type 301) and one 3-thread overlock (stitch type 504), both threaded and fitted with the maker's standard attachments. **Materials:** medium-weight cotton poplin specimens (40 cm × 20 cm), sewing thread cones of 5,000 m, machine needles (DB×1 for lockstitch, DC×27 for overlock), stopwatch, stitch counter or ruler, machine manual, screwdriver and cleaning brush.

3. Anatomy of the lockstitch machine

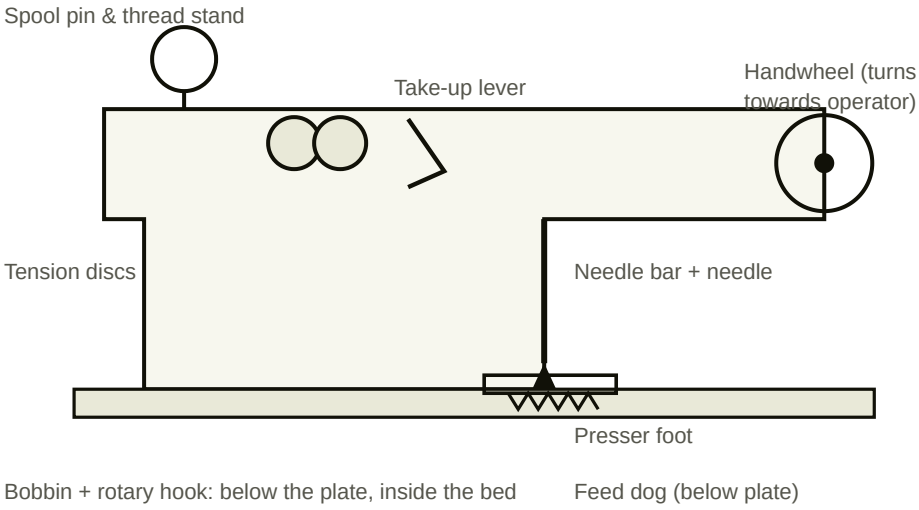


Figure 1 — Single-needle lockstitch machine head, working parts. The bed, arm and head form the frame; everything that makes the stitch hangs from them.

PART	FUNCTION IN THE STITCH CYCLE
Needle bar and needle	Carries the needle down through the fabric; as the needle rises about 1.8–2.2 mm from bottom dead centre it throws a loop of needle thread.
Rotary hook and bobbin	The hook point enters the needle loop, carries it around the bobbin case so the needle thread wraps the bobbin thread, then releases it — the interlacing that forms stitch 301.
Thread take-up lever	Supplies thread for the loop, then pulls the stitch tight into the fabric; its stroke is the visible heartbeat of the machine.
Tension discs and spring	Meter the needle thread against the bobbin-thread reference tension; their balance buries the knot mid-ply.
Feed dog	Rises, grips the fabric against the presser foot, carries it back exactly one stitch length, drops and returns — an elliptical four-motion feed.
Presser foot and bar	Holds the fabric against the feed dog with adjustable spring pressure; lifted by knee lifter or lever for positioning.
Stitch-length regulator	Sets the feed-dog throw, and so the stitch length; read as stitches per cm or per inch on the dial's scale.
Motor and treadle	Servo or clutch motor under the table; treadle controls start, speed and stop, the heel movement operates the thread trimmer on automatic machines.

4. The overlock machine, in brief

The overlock replaces the bobbin with two **loopers**: the upper looper carries its thread across the fabric edge and the lower looper passes its thread through the needle loop and the upper looper's loop, so three threads wrap the cut edge (stitch 504). Two **knives** — a fixed lower knife and a moving upper knife — trim the edge just ahead of the needles, which is why overlocked edges are always clean-cut. Most industrial overlocks add **differential feed**: two feed dogs whose relative throw can be set to stretch or gather the fabric as it sews, the control that prevents wavy seams on knits and puckering on fine wovens.

5. Observed performance: worked verification

(a) Feed speed from rated rpm. The maker's plate rates the lockstitch at 5,000 stitches per minute (spm) maximum. At the study setting of 4 stitches/cm the stitch length is:

$$\text{stitch length} = 10\ \text{mm} \div 4 = 2.5\ \text{mm}$$

so the theoretical fabric feed speed at full rpm is 5,000 × 2.5 mm = 12,500 mm/min = **12.5 m/min**. In practice the operator rarely holds full speed — start/stop, turning and handling consume most of the cycle — which is exactly why handling, not machine speed, dominates SMV (Study 5).

(b) Thread consumption. For stitch 301 the total thread consumed is about **2.5 times the seam length** (roughly 1.35 parts needle thread and 1.15 parts bobbin thread, varying with fabric thickness and tension). The specimen seams total 500 cm, so:

$$\text{thread per garment} = 500\ \text{cm} \times 2.5 = 1{,}250\ \text{cm} = 12.5\ \text{m};\ \text{with 10\% allowance} = 13.75\ \text{m}$$

A 5,000 m cone therefore sews 5,000 ÷ 13.75 = **363 garments** per cone position — the figure the store uses when converting an order quantity into cone purchases, per thread position.

(c) Stitch-density check. Sew 10 cm on the specimen, count the stitches along a ruled 5 cm centre stretch, and multiply by 2. A reading of 20 stitches in 5 cm confirms the 4/cm setting; a reading of 18 means the feed throw has drifted to 3.6/cm (stitch length 2.78 mm) and the regulator needs resetting before any production specimen is sewn.

6. Procedure summary

1. With the machine off, identify and log every part of Figure 1 on the actual machine; state each function aloud to the partner.
2. Trace one full stitch cycle by turning the handwheel towards you: needle descent, loop formation, hook entry, thread wrap, take-up pull-up, feed movement.

3. Repeat the identification on the overlock: loopers, knives, differential-feed control.
4. Sew the specimen seams; perform the density count and thread-consumption calculation of §5 and record all values in the study report.

Precautions

- Switch off before any work near the needle, hook or knives; turn the handwheel only towards the operator.
- Keep fingers behind the needle guard line; use tweezers, never fingers, to position small pieces under the foot.
- Report a bent or burred needle at once — it damages the hook and the fabric, and needle fragments are a critical defect in finished garments.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Walk through one lockstitch cycle: what does the hook do, and when?
2. A machine is set to 5 stitches/cm and runs at 4,000 spm. What is the stitch length and the theoretical feed speed?
3. Why does the overlock need knives when the lockstitch does not?
4. A style has 620 cm of lockstitch seam per garment. Estimate the thread per garment with allowance, and the garments per 5,000 m cone.

Answer key

1. As the needle rises 1.8–2.2 mm from its lowest point it forms a loop; the hook point enters the loop and carries the needle thread around the bobbin case, interlacing it with the bobbin thread; the take-up lever then pulls the stitch tight as the feed dog advances the fabric one stitch length.
2. Stitch length = 10 ÷ 5 = 2.0 mm. Feed speed = 4,000 × 2.0 mm = 8,000 mm/min = 8.0 m/min.
3. The overlock stitch wraps the fabric edge itself, so the edge must be trimmed true in the same pass; the moving and fixed knives cut it immediately ahead of the needles. The lockstitch sews inside the fabric and needs no trimmed edge.
4. Thread = 620 × 2.5 = 1,550 cm = 15.5 m; with 10% allowance 17.05 m. Cones: 5,000 ÷ 17.05 ≈ 293 garments per cone position.